



B# FoodPanda Clone – Database Project Proposal

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1. Project Overview

Objective: Design and implement a complete relational database for a food delivery platform (inspired by FoodPanda). The system handles customers, restaurants, riders, orders, payments, promotions, reviews, and logistics – all with full referential integrity and security features.

Scope: 22 tables, 50+ attributes, primary/foreign keys, normalization up to 3NF, sample data, and ready-to-run SQL scripts for Microsoft SQL Server.

Unique Features: Cyber-secure user table (MFA), digital evidence logs, heatmap logs for customer activity, and a complete order lifecycle tracking.

2. Three-Level Database Architecture



Conceptual Level

Entities: Customer, Restaurant, Rider, Order, Payment, MenuItem.

Relationships: "Customer places Order", "Rider delivers Order". No technical details.



Logical Level

Tables, attributes, data types, primary/foreign keys, constraints. 22 tables fully normalized (3NF).



Physical Level

Indexes (clustered on PK), storage, security (User_Security table), audit logs, encryption recommendations.



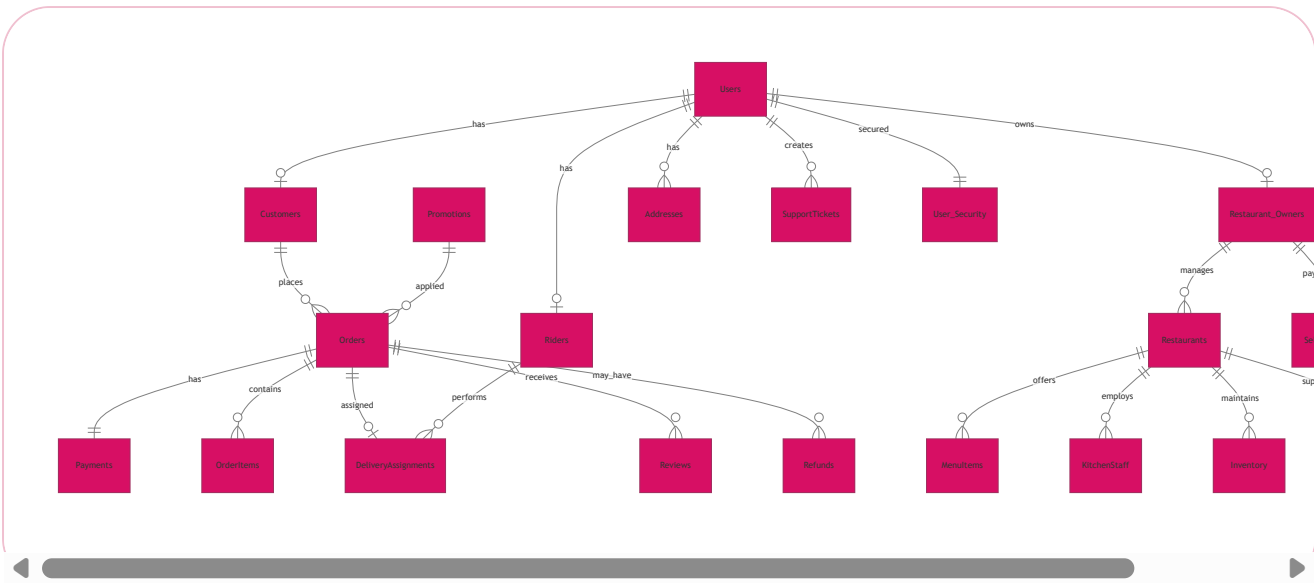
3. Complete Table Inventory (22 Tables)

Below is the complete list of tables with primary keys (🔑 PK) and foreign keys (🔗 FK).

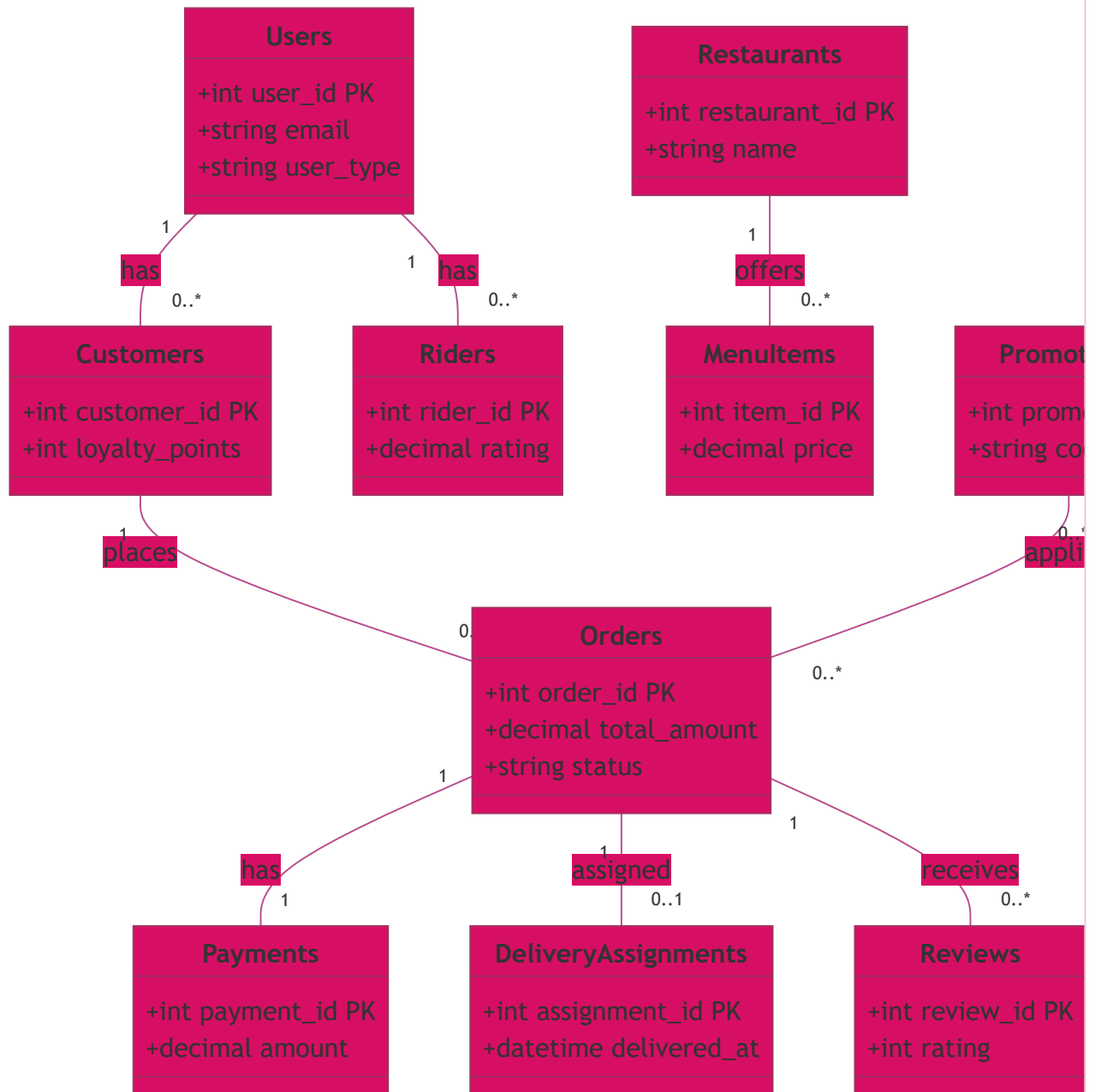
Table Name	Primary Key	Foreign Key(s)	Description
Users	user_id 🔑	-	Base authentication table
Customers	customer_id 🔑	user_id 🔗	Customer-specific details
Restaurant_Owners	owner_id 🔑	user_id 🔗	Restaurant owners
Riders	rider_id 🔑	user_id 🔗	Delivery riders
Restaurants	restaurant_id 🔑	owner_id 🔗	Restaurant information
MenuItems	item_id 🔑	restaurant_id 🔗	Food items with price
Orders	order_id 🔑	customer_id, restaurant_id 🔗	Main order header
OrderItems	order_item_id 🔑	order_id, item_id 🔗	Line items of orders
Payments	payment_id 🔑	order_id 🔗	Transaction records
DeliveryAssignments	assignment_id 🔑	order_id, rider_id 🔗	Rider assignment & timeline
Promotions	promo_id 🔑	-	Discount codes
Reviews	review_id 🔑	order_id 🔗	Ratings and comments
KitchenStaff	staff_id 🔑	restaurant_id 🔗	Employees working in kitchens
Inventory	inventory_id 🔑	restaurant_id 🔗	Stock management
SupportTickets	ticket_id 🔑	user_id 🔗	Customer support tickets
Refunds	refund_id 🔑	order_id 🔗	Refund requests
SellerWallet	wallet_id 🔑	owner_id 🔗	Payout balance for restaurant owners
HeatmapLogs	log_id 🔑	-	Customer activity coordinates
SystemHealth	health_id 🔑	-	Platform monitoring logs
GroceryItems	grocery_id 🔑	restaurant_id 🔗	Non-food items (groceries)
Addresses	address_id 🔑	user_id 🔗	Shipping addresses
User_Security	user_id 🔑 🔗	user_id 🔗	MFA, last login, security flags

Total: 22 tables

4. Entity-Relationship Diagram – Crow's Foot Notation



5. UML Class Diagram (Simplified)



6. Normalization & Integrity

- ✓ All tables are in **Third Normal Form (3NF)** – no transitive dependencies, no repeating groups.
- ✓ Primary keys defined as identity columns.
- ✓ Foreign keys enforce referential integrity (CASCADE DELETE where appropriate, e.g., OrderItems → Orders).
- ✓ Check constraints for user_type, ratings, and status fields.
- ✓ Triggers can be added for audit logging (User_Security updates).

7. Conclusion & Future Roadmap

This database design provides a complete, scalable, and secure backend for a food delivery platform. It meets all university project requirements and can be extended into a full-stack application.

- ♦ Real-time order tracking using WebSockets (SignalR)
- ♦ AI-based demand forecasting (ML integration)
- ♦ Mobile app for customers & riders (React Native)
- ♦ Blockchain payment (GCB Coin integration)
- ♦ Advanced analytics dashboard with Power BI

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